# PROJECT: AWS EC2 Auto Deployment using Docker

**Services:** EC2, Jenkins, Docker Project

**Goal:** automate the process of building, storing, and deploying an application using a CI/CD pipeline so that whenever code is updated, the application is automatically deployed on an AWS EC2 server using Docker containers.

**Skills:** CI/CD Pipeline Implementation, Containerization, Docker Image Registry management.

## Definition:

AWS EC2 Auto Deployment using Docker is a DevOps approach that automates application deployment in the cloud. It uses Amazon EC2 for hosting, Docker for containerization, and Jenkins for CI/CD automation. When code is pushed to GitHub, Jenkins builds a Docker image and deploys it automatically on EC2. This process reduces manual work, ensures consistency, and enables faster and reliable deployments.

## Scope of the Project:

The scope of this project is to design and implement an automated deployment system using cloud and DevOps technologies. It involves integrating Amazon EC2, Docker, and Jenkins to build, store, and deploy applications through a CI/CD pipeline. The project includes creating Docker images, pushing them to a registry like Docker Hub, and automatically deploying containers on EC2 instances. It aims to improve deployment speed, reduce human errors, and ensure consistency. The solution can be extended for scalable cloud applications and microservices architecture.

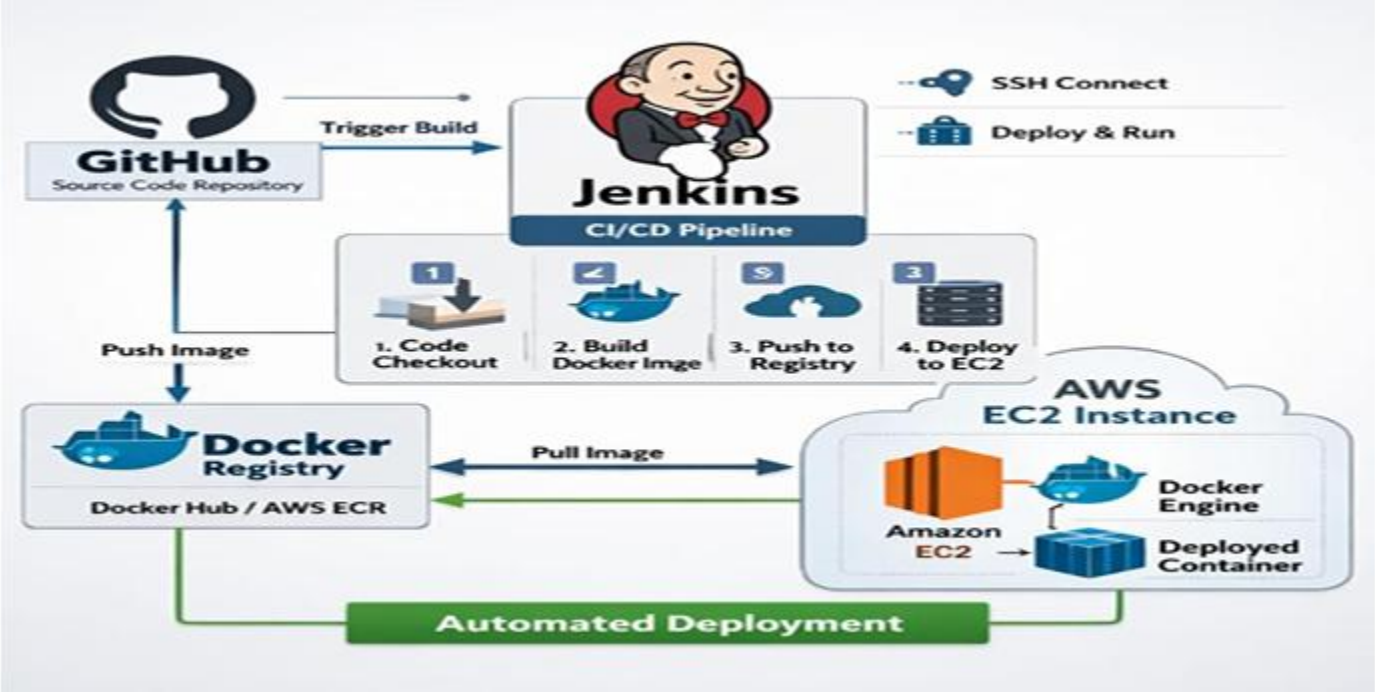
### Methodologies:

1. The project follows the **CI/CD (Continuous Integration and Continuous Deployment)** approach for automation.
2. Code changes pushed to GitHub trigger automated build and deployment using Jenkins.
3. The application is containerized using Docker to ensure consistency across environments.
4. Docker images are stored in a registry like Docker Hub for easy access and version control.
5. The application is deployed automatically on Amazon EC2 without manual intervention.
6. This methodology reduces errors, speeds up deployment, and supports scalable cloud applications.

## Services Used in this Project:

1. Amazon EC2 – Provides a virtual server in the cloud to host and run the application and Docker containers.
2. Jenkins – Acts as the CI/CD tool to automate building, testing, and deployment of the application.
3. Docker – Used for containerizing the application to ensure it runs consistently across environments.
4. Docker Hub – Stores and manages Docker images so they can be easily pulled and deployed.
5. GitHub – Hosts the source code repository and triggers the CI/CD pipeline when changes are pushed.

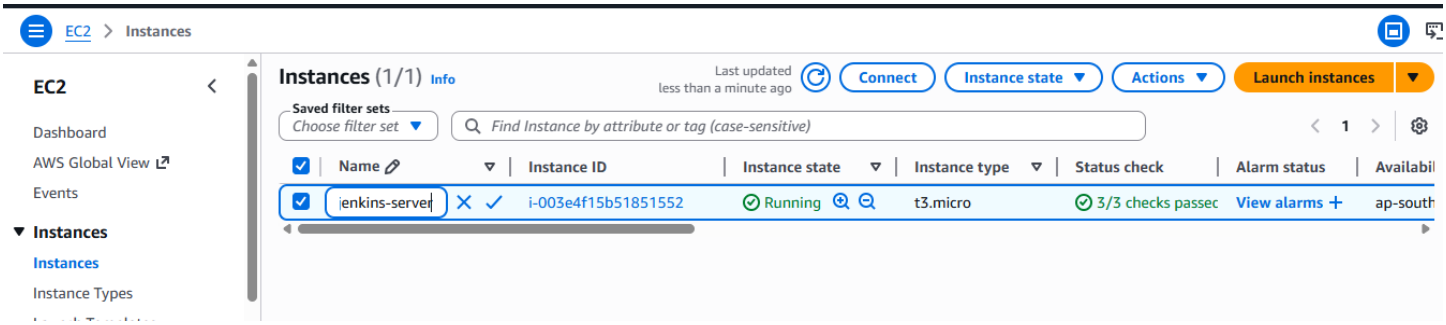
## Architecture Diagram:




## Implementation and Execution of AWS EC2 Auto Deployment using Docker.

**STEP 1:** Create AWS EC2 Instance

- Launch an Amazon EC2 instance in AWS.
- Choose Amazon Linux AMI.
- Configure security group (ports 22, 8080, 8081).




**STEP 2:** Install Required Software.

```
[root@ip-172-31-44-97 my-app]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-44-97 my-app]# jenkins --version
2.541.3
[root@ip-172-31-44-97 my-app]#
```

**STEP 3:** Create Dockerfile


**STEP 4:** Create a Jenkinsfile to define pipeline stages.

Pipeline stages:

➢ Code Checkout

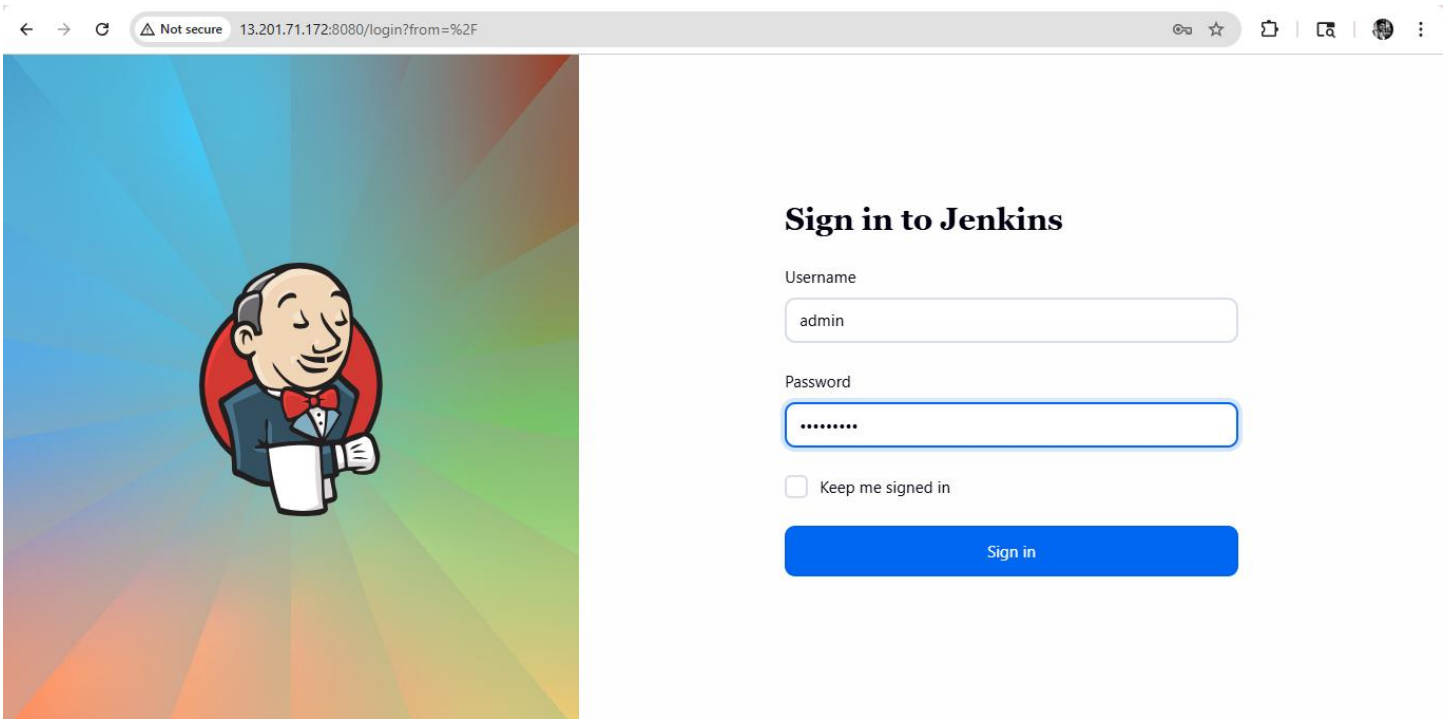
➢ Build Docker Image

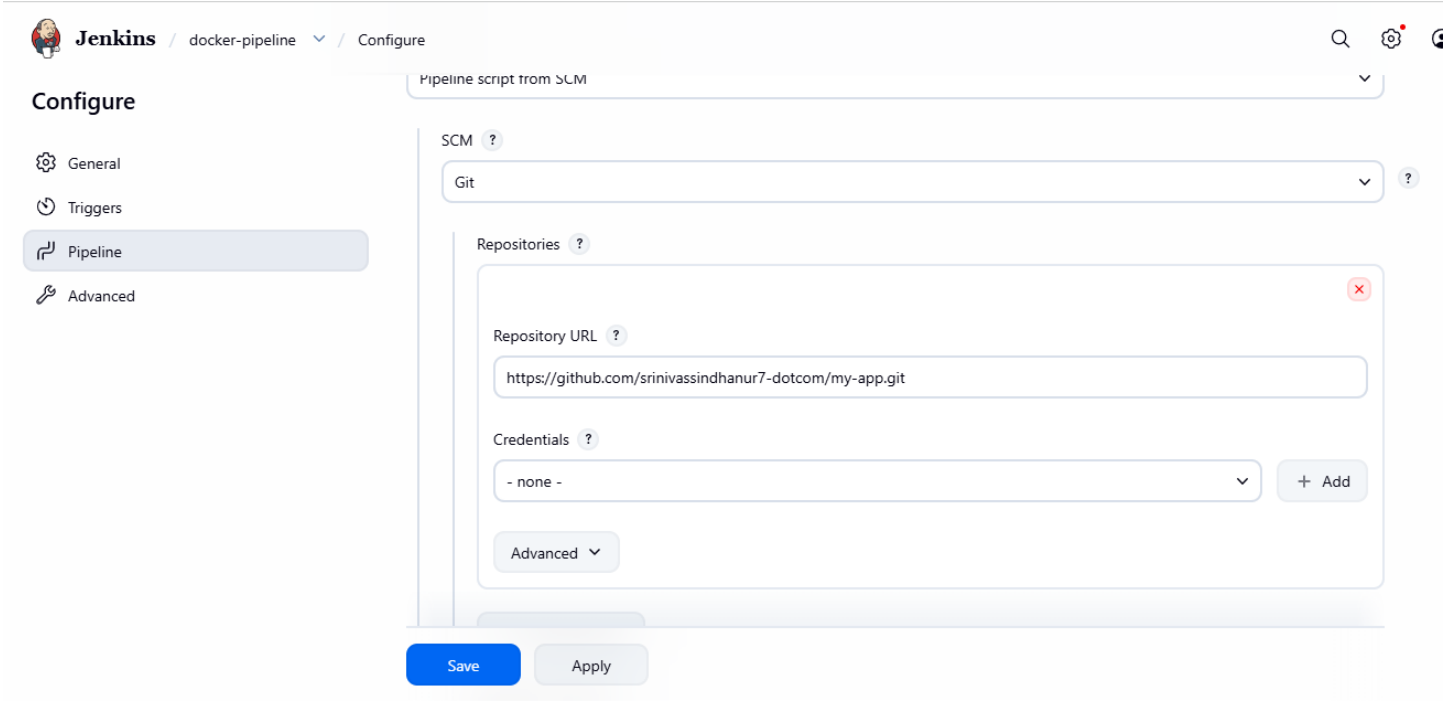
➢ Push Image to Docker Registry

➢ Deploy Container

```
  GNU nano 8.3                                  Jenkinsfile
pipeline {
    agent any

    stages {

        stage('Checkout Code') {
            steps {
                git branch: 'main', url: 'https://github.com/srinivassindhanur7-dotcom/my-app.git'
            }
        }

        stage('Build Project') {
            steps {
                sh 'mvn clean package'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t my-app .'
                                [ Read 32 lines ]
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was
```

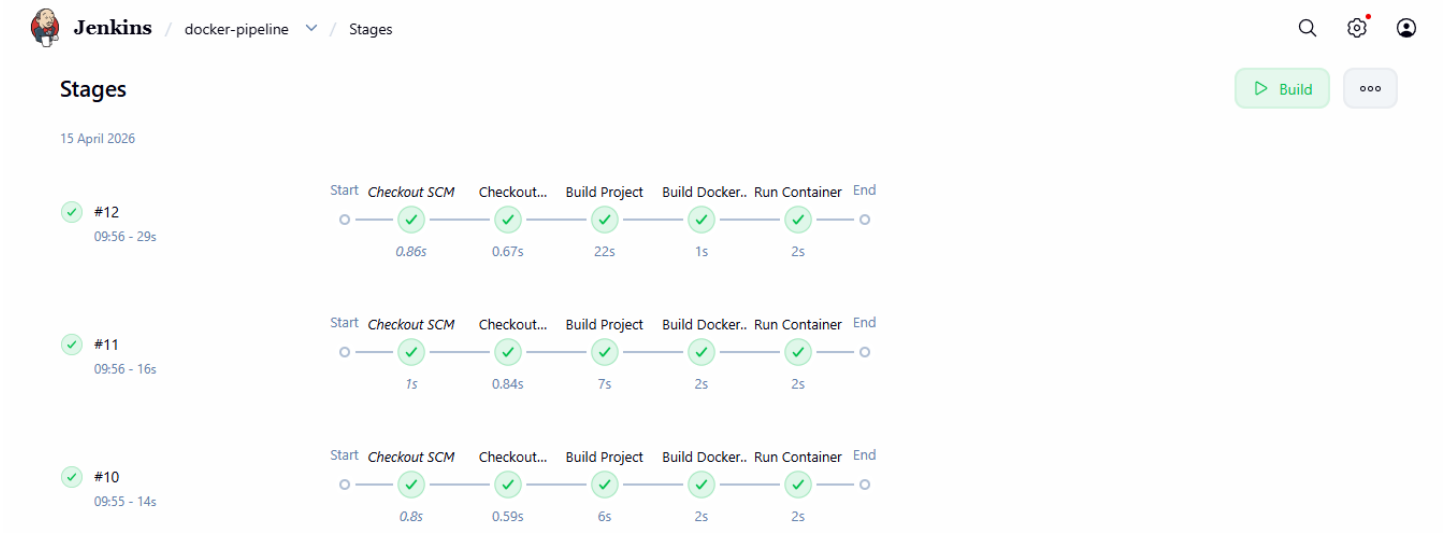
**STEP 4:** Configure Jenkins.

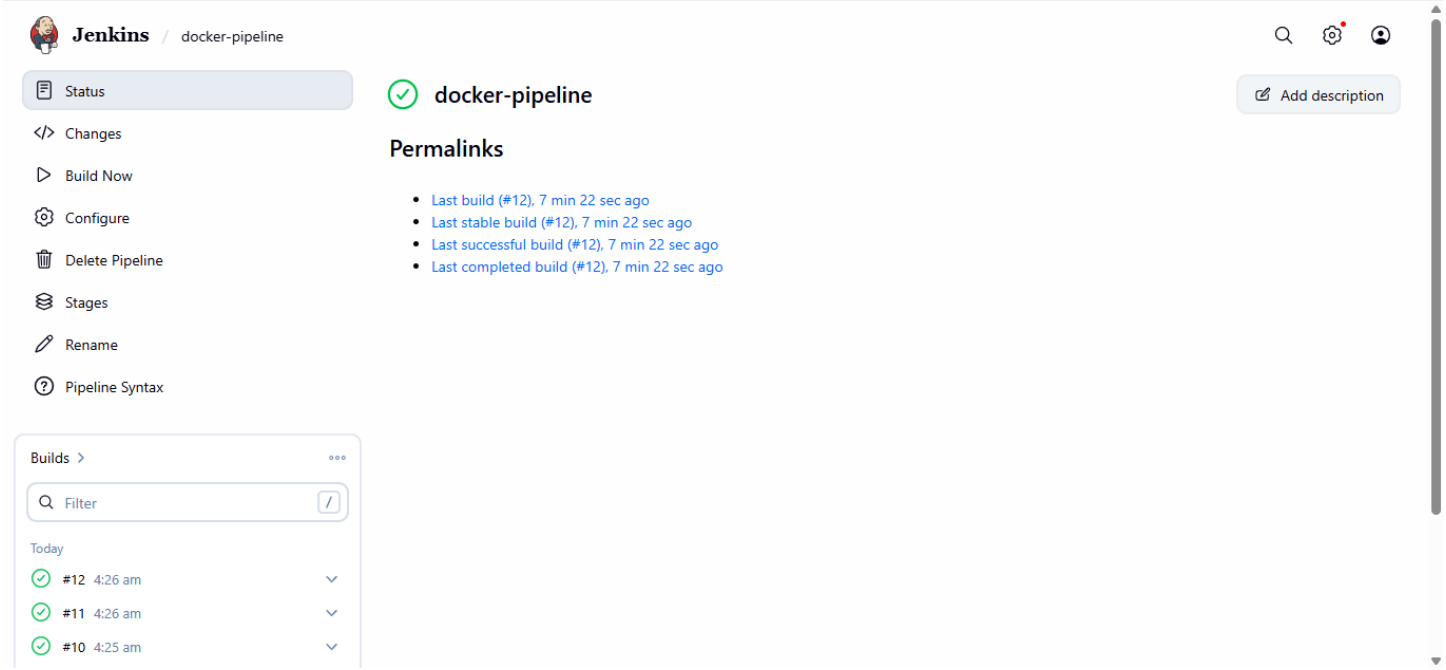




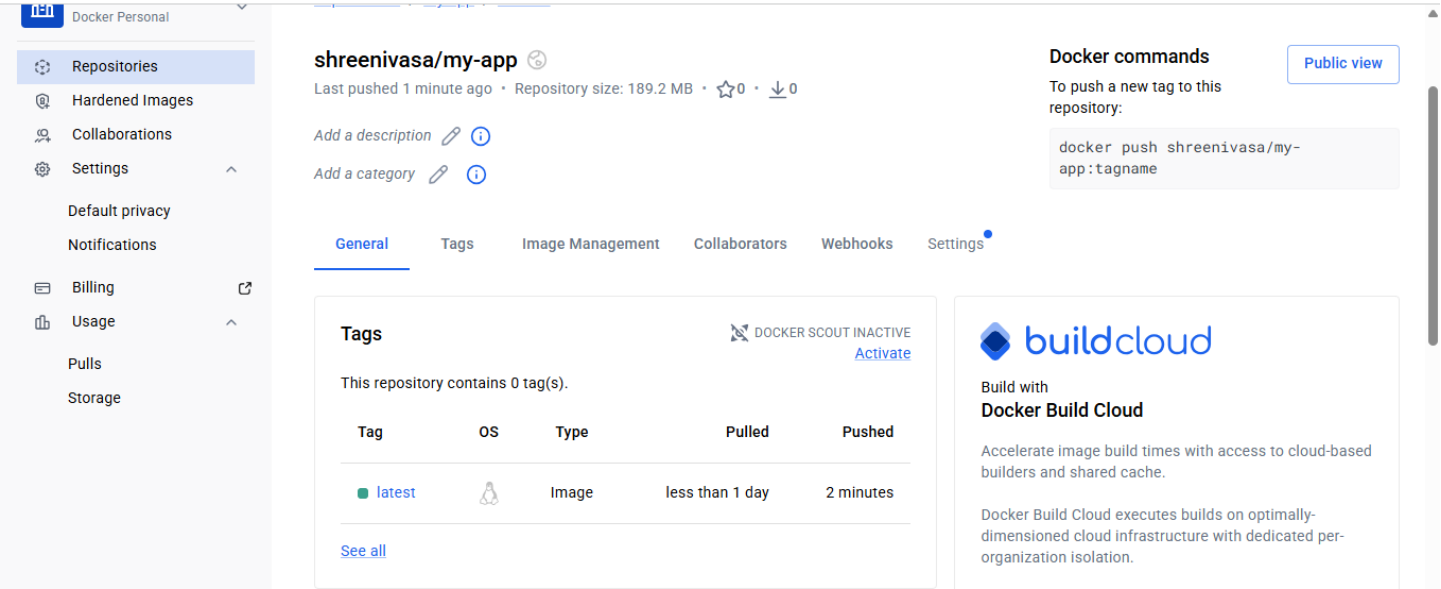

**STEP 5**: Built the image successfully in Jenkins

**STEP 8:** Successfully built the pipeline job.



**STEP 7:** Image is pushed to the DockerHub.




**Conclusion:** The AWS EC2 Auto Deployment using Docker project successfully demonstrates how cloud computing, containerization, and CI/CD automation can be integrated to streamline application deployment. By using Amazon EC2, Docker, and Jenkins, the project automates the process of building, storing, and deploying applications. The CI/CD pipeline ensures that any changes made in GitHub are automatically built into Docker images and deployed on the EC2 server. This approach reduces manual effort, minimizes deployment errors, and improves development efficiency. Overall, the project highlights the importance of DevOps practices in modern cloud-based application deployment.